

Sanjana Peter

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Professional Summary

Senior Software Engineer with 8 years of professional experience specializing in **scalable backend application architecture, Node.js/TypeScript development, and AI/LLM integration**. Strong hands-on experience designing RESTful APIs, data processing pipelines, and relational database structures. Proven track record executing backend optimizations, deploying localized network configurations, and developing active Retrieval-Augmented Generation (RAG) workflows utilizing cloud-integrated services.

Experience

iSpace Software Solutions

Remote, USA

Software Engineer

Jun 2025 – Present

- Architected a scalable ETL pipeline using **NestJS** and **TypeORM** to ingest, process, and normalize data from enterprise EdTech APIs into a PostgreSQL database.
- Quickly onboarded into an active engineering cycle, unblocking feature delivery timelines and providing technical support across the team to ensure smooth production features.
- Engineered a personalized recommendation system using advanced **PostgreSQL** query optimization and event-driven application logic, boosting user learning engagement metrics by 50%.

Marist College

Poughkeepsie, NY

Student Network Technician

Sep 2023 – May 2025

- Designed and implemented physical network infrastructure, including the deployment, cable management, and hardware labeling of a dedicated, in-house LAN server room.
- Provisioned, managed, and troubleshooted institutional network switches, routers, and access points; audited network performance metrics to maintain high security and minimal latency boundaries.
- Collaborated on localized infrastructure upgrades and network-related internal tooling automation scripts using Bash, improving IT systems maintenance workflows.

Elptico IT Solutions Private Limited

Hyderabad, India

Senior Software Engineer (Backend)

Oct 2020 – Jul 2023

- Designed and delivered a critical, HIPAA-compliant healthcare enterprise platform utilizing **Node.js services** and **DynamoDB** to implement secure, member-centric data workflows.
- Developed scalable backend application components integrated with cloud services including **AWS Lambda** and **API Gateway** for secure endpoint routing.
- Integrated SAML 2.0 Single Sign-On (SSO) for enterprise security using Azure Active Directory, establishing robust role-based access control (RBAC) across applications.
- Optimized complex transactional patterns and index strategies on **PostgreSQL**, accelerating backend order processing and inventory synchronization tasks.
- Mentored and guided a team of 12 backend developers, instituting rigorous code review standards that decreased overall production regression defects by 15%.
- Implemented an automated test suite using **Jest**, achieving 80% code coverage across core data ingestion modules to secure continuous integration stability.

Techlet Solutions Pvt. Ltd.

Delhi, India

Software Engineer

Jul 2016 – Oct 2020

- Developed and maintained highly available RESTful services and real-time messaging layers using **Node.js**, PHP, and relational databases supporting concurrent users.
- Utilized cloud monitoring metrics and logging systems to identify API latencies, ensuring consistent application availability and system performance.
- Built data-driven backend architectures and analytical pipelines leveraging MySQL, MongoDB, and Elasticsearch to power near real-time business reporting panels.
- Authored utility scripts and serverless functions using Node.js and Python to streamline event-driven background automation tasks.

Active Engineering Projects

AI Healthcare Chatbot Application

Lead Architect & Developer

Node.js, TypeScript, LangChain, Vector Databases, Python

2025 – Present

- Designing and building a comprehensive **Retrieval-Augmented Generation (RAG)** chatbot application engineered for symptom-based conversational routing and context lookup.
- Developing data ingestion workers in Node.js to clean and format unstructured medical texts, establishing ingestion pathways for vector embedding and semantic search storage.
- Implementing prompt engineering constraints and context matching guards using LLM frameworks to ensure high accuracy and mitigate hallucination vectors.

BLE Tracking & Machine Learning Research

Python, Scikit-Learn, Pandas, NumPy, BLE Infrastructure

- Designed and implemented a predictive machine learning model utilizing **Python** and **Scikit-Learn** to analyze Bluetooth Low Energy (BLE) signal patterns for real-time indoor tracking.
- Engineered custom data cleaning pipelines using **Pandas** and **NumPy** to handle noisy RSSI time-series data, eliminating signal anomalies and improving predictive baseline consistency.
- Evaluated classification algorithms to optimize location accuracy boundaries, successfully mapping physical device presence coordinates based on telemetry data logs.

Lead ML Developer

2024 – 2025

CareerPilot-Apply Platform

TypeScript, Node.js, Next.js, PostgreSQL, AI Development Tools

- Constructing a web application dashboard designed to centralize status updates, pipeline tracking, and metrics management for job applications.
- Leveraging advanced generative AI development workflows (GitHub Copilot) to accelerate core feature prototyping and boilerplate layout setups.
- Modeling a normalized PostgreSQL relational database schema to ensure robust data persistence, relationship integrity, and application state tracking.

Creator & Core Developer

2025 – Present

Technical Skills

Languages & Ecosystems: TypeScript, JavaScript (ES6+), Node.js, NestJS, Express.js, Python, PHP, SQL, Bash

Cloud Integration: AWS (Lambda, API Gateway, CloudWatch, IAM, ECS), Docker, CI/CD Concepts

Databases & Caching: PostgreSQL, DynamoDB, Vector Databases, MongoDB, MySQL, Redis, Elasticsearch

Networking & Security: LAN Room Topology, Switch/Router Configuration, SAML 2.0, OAuth, RBAC, HIPAA Compliance

AI & Machine Learning: Retrieval-Augmented Generation (RAG), Vector Embeddings, Scikit-Learn, Pandas, NumPy, Prompt Engineering

Education

Marist College

Master of Science in Information Systems Management; GPA: 3.95/4.00

Poughkeepsie, NY

2025

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology in Computer Science

Lucknow, India

2016